

# CLIENT BRIEF

## Emergency Department Operational Performance & Patient Experience Analytics

**Client:** St. Jude's Health System – Operational Excellence Division

**Project Duration:** 4 Weeks

**Issue Date:** October 2024

### 1. ABOUT ST. JUDE'S HEALTH SYSTEM

St. Jude's is a leading metropolitan healthcare provider dedicated to delivering high-quality emergency care. Our Emergency Departments (ED) serve as the frontline of our healthcare system, managing thousands of acute cases annually.

Our Operational Excellence Division is currently focused on "Patient-First Analytics." We aim to move beyond simple record-keeping toward a proactive data culture where real-time insights dictate staffing levels, resource allocation, and clinical workflow improvements to ensure every patient receives timely and compassionate care.

### 2. PROJECT CONTEXT

**The Emergency Room Ecosystem** Emergency Rooms operate in a high-pressure environment where time-to-care is the most critical metric. The patient journey typically follows a set path:

- **Triage:** Initial assessment of severity.
- **Wait Time:** The interval between arrival and seeing a clinician.
- **Referral/Treatment:** Coordination with specialised departments (Cardiology, Orthopedics, etc.).
- **Disposition:** The final decision to either "Admit" the patient to the hospital or "Treat and Release."

**Current Challenges** Internal reviews suggest that while our clinical outcomes remain strong, our operational efficiency is fluctuating. We are seeing "bottleneck" periods where wait times spike, and these periods correlate strongly with a drop in our Patient Satisfaction Scores. To address this, we need a transparent, interactive reporting suite that allows different levels of hospital leadership to see exactly where the system is under strain.

### 3. THE BUSINESS CHALLENGE

St. Jude's has identified a disconnect between volume management and patient experience. With over 9,000 annual visits, the ER is experiencing "moderate" wait times (avg. 35.3 mins) and "below-target" satisfaction scores (avg. 4.99/10).

**Specific challenges include:**

- **Peak Demand Surges:** Preliminary data shows significant volume spikes on Mondays and late-night hours (11 PM), but staffing rosters have not been updated to match these trends.
- **Departmental Pressure:** High referral rates to General Practice and Orthopedics are creating "exit blocks" in the ER.
- **Demographic Sensitivity:** We need to ensure that care is equitable and high-quality across all age groups and ethnic backgrounds, particularly within our largest patient demographic (20–39 years).

## 4. PROJECT OBJECTIVES

We require a Senior Data Analyst to develop a Power BI solution that will:

- **Primary Objective:** Create a 3-tier dashboard suite (Consolidated, Monthly, and Patient Details) to monitor ER performance.
- **Uncover Insights:** Identify the exact hours and days where wait times exceed our 30-minute internal benchmark.
- **Staffing Alignment:** Provide a heatmap of patient arrivals to assist HR in roster optimization.
- **Actionable Recommendations:** Suggest targeted interventions for high-referral departments.

### Key Analytical Questions:

1. What is the correlation between **Wait Time** and **Patient Satisfaction**?
2. Which **Days of the Week** and **Hours of the Day** represent the highest risk for overcrowding?
3. How do **Departmental Referral** patterns impact the speed of patient disposition?
4. Are there specific **Demographic Groups** (Age/Race) that report lower satisfaction scores?
5. What is the current ratio of **Admissions vs. Releases**, and how does this affect bed capacity planning?

## 5. AVAILABLE DATA ASSETS

We are providing a validated dataset containing 9,216 patient records.

Dataset: Hospital ER\_Data.csv

### The dataset includes:

- **Patient Metrics:** Satisfaction scores (1-10), Wait times (minutes).
- **Temporal Data:** Full admission timestamps.
- **Demographics:** Patient Age, Gender, and Race.
- **Clinical Data:** Admission flags (True/False) and Departmental Referrals.

## 6. SCOPE OF WORK

### 6.1 Data Engineering & Modeling

- Clean and transform the CSV data using Power Query.

- Create a Date/Time dimension to allow for "Hour of Day" and "Day Name" analysis.
- Develop DAX measures for KPIs: Avg Wait Time, Avg Satisfaction, Admission Rate %, and Total Patient Count.

## 6.2 Data Visualisation

### The 3 Required Dashboards:

1. **Consolidated View (Executive):** High-level summary of KPIs, admission trends, and referral distribution.
2. **Monthly View (Operational):** Temporal analysis, including the "Demand Heatmap" for staffing.
3. **Patient Details (Granular):** A searchable drill-through page with filters for Race, Age, and Gender.

## 7. DELIVERABLES

- **A Power BI File (.pbix):** Containing the three interactive dashboards.
- **Executive Summary Report:** A brief (max 500 words) outlining the three biggest "So What" findings and recommended actions.

## 8. SUCCESS CRITERIA

"Good" looks like:

- **Clarity:** An executive can understand the ER's health in under 30 seconds of viewing the Consolidated page.
- **Accuracy:** Visuals correctly reflect the 35.3-minute wait time and 4.99 satisfaction averages.
- **Interactivity:** Using a slicer on the "Patient Details" page should instantly update the demographic breakdown.